Homework #03

CMOS Course

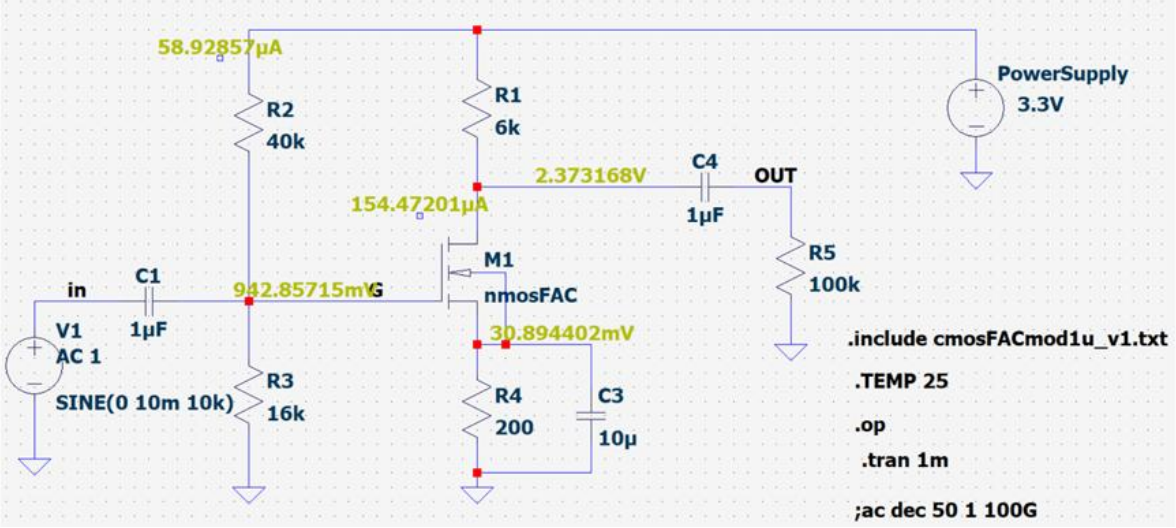
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Deadline: July 19, 2026

Analyze the direct current (DC) and alternating current (AC) behavior of a small-signal amplifier using the 2N7000 MOSFET in LTspice.

Preliminary instructions for setting up the circuit in LTspice:

- Use the schematic shown in the figure below. Replace the nmosFAC with the generic component labeled “nmos.”
- Right-click on the transistor, select “Pick New Transistor,” and choose the 2N7000.



Model 2N7000

```
.SUBCKT 2N7000 20 10 30
Rg 10 1 1
M1 2 1 3 3 DMOS L=1u W=1u
.MODEL DMOS NMOS (VTO={2.1*{-0.0016*TEMP+1.04}} KP=0.35 THETA=0.086
+ VMAX=2.2E5 LEVEL=3)
Cgs 1 3 60p
Rd 20 4 0.3 TC=0.0075
Dds 3 4 DDS
.MODEL DDS D(BV={60*{0.00072*TEMP+0.982}} M=0.36 CJO=23p VJ=0.8)
Dbody 3 20 DBODY
.MODEL DBODY D(IS=1.4E-13 N=1 RS=40m TT=100n)
Ra 4 2 0.4 TC=0.0075
Rs 3 5 10m
Ls 5 30 .5n
M2 1 8 6 6 INTER
E2 8 6 4 1 2
.MODEL INTER NMOS(VTO=0 KP=10 LEVEL=1)
Cgdmax 7 4 85p
Rcgd 7 4 10meg
Dgd 6 4 DGD
Rdgd 4 6 10meg
.MODEL DGD D(M=0.53 CJO=85p VJ=0.12)
M3 7 9 1 1 INTER
E3 9 1 4 1 -2
.ENDS
```

Phase 1: DC Operating Point Analysis

The objective of this phase is to ensure the transistor is turned on and properly biased within its linear amplification zone (the saturation region).

- Theoretical Calculations: Using the equations for a MOSFET in saturation, calculate the theoretical gate voltage (V_G), drain current (I_D), and drain-source voltage (V_{DS}). (Note: Use the standard threshold voltage and conduction parameters found in the 2N7000 datasheet model).

- LTspice Simulation: Build the circuit using the nmos symbol, assign it the 2N7000 model, and run a DC operating point simulation using the .op command.

- Present a clean, structured comparative table showing your hand calculations alongside the simulated LTspice results for V_G, I_D, and V_{DS}.

Phase 2: Transient Response (Time-Domain Analysis)

The objective of this phase is to observe how the circuit dynamically processes and amplifies a time-varying small signal.

- Source Configuration: Set the input voltage source to a sine wave (SINE) with a peak amplitude of **10 mV** and a frequency of **10 kHz**.

- LTspice Simulation: Configure and run a transient analysis simulation using the .tran command. Set the stop time to display exactly 4 full cycles of the waveform on your screen (total time: **4 ms**).

- Plot the input signal and the amplified output signal simultaneously on the same graph window.

  Measure the peak-to-peak voltage of both the input ($V_{in,pp}$) and the output ($V_{out,pp}$), then calculate the midband voltage gain **(**A_v**)** using the following expression:

$$A_v = \frac{V_{out,pp}}{V_{in,pp}}$$

Submission Guidelines:

Format**:** You must export or scan your final solution into a single **PDF format**.

Platform: Upload the PDF file directly to our **Google Classroom** assignment slot before the deadline.